



Implementation of DHCP, DNS, Active Directory, and Group Policy for Network Management

Group : 3

Names : Aldi Syarif Alhakim
Aleazuna M. Nadhif Pohan
Muhamad Ihsad Rasyad

Class : 2CS1

CEP CCIT

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PROJECT ON

Windows Server for Network Management

Developed by

- 1. Aldi Syarif Alhakim**
- 2. Alezuna M. Nadhif Pohan**
- 3. Muhamad Ihsan Rasyad**

Windows Server for Network Management

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Name of Faculty : Tri Agus Riyadi, S.Kom., M.T.

Names of Developer :

1. Aldi Syarif Alhakim
2. Alezuna M. Nadhif Pohan
3. Muhamad Ihsan Rasyad

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CERTIFICATE OF ORIGINALITY

This is to certify that the project report titled "Implementation of DHCP, DNS, Active Directory, and Group Policy for Network Management" is an original work completed by Aldi Syarif Alhakim, Alezuna M. Nadhif Pohan and Muhamad Ihsan Rasyad. This project has been submitted in partial fulfillment of their course requirement at the National Institute of Information Technology (NIIT). The project report has been prepared under our guidance and supervision, and it is ensured that the work presented in this report is the result of the individual efforts of the aforementioned students. The contents of this report have not been submitted to any other institution or organization for the award of any degree, diploma, or other similar recognition. Author acknowledge that the ideas, designs, and implementations presented in this project report are the intellectual properties of the students mentioned above. Any use or reproduction of this work must give proper credit to the original authors. Author hereby endorse the authenticity and originality of the work presented in this project report and confirm that it meets the academic standards and requirements set forth by the National Institute of Information Technology (NIIT).

Coordinator:

Tri Agus Riyadi, S.Kom., M.T.

BACKGROUND

In the modern era of information technology, the management of computer networks within an organization has become increasingly complex and critical. Organizations, whether educational institutions, companies, or government agencies, require a reliable, centralized, and efficient network infrastructure to support their daily operational activities.

Windows Server, developed by Microsoft, is one of the most widely used server operating systems in enterprise environments. It provides a comprehensive set of services that enable network administrators to manage users, devices, and resources in a structured and scalable manner. Among the most essential services offered are Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), Active Directory Domain Services (AD DS), and Group Policy Objects (GPO).

DHCP plays a fundamental role in automating the assignment of IP addresses to client devices within a network, eliminating the need for manual configuration and reducing the risk of address conflicts. DNS, on the other hand, is responsible for translating human-readable domain names into IP addresses, enabling seamless communication between devices across the network.

Active Directory Domain Services (AD DS) serves as the backbone of centralized identity and access management, allowing administrators to create and manage user accounts, computers, and organizational units within a domain environment. Building upon AD DS, Group Policy Objects (GPO) provide a powerful mechanism for enforcing security settings, software configurations, and user restrictions across all machines and users within the domain.

This paper aims to explore and demonstrate the configuration of these four essential services — DHCP, DNS, AD DS, and GPO — within a Windows Server environment, providing a practical understanding of how they work together to build a functional and secure network infrastructure.

SYSTEM ANALYSIS

This paper presents the configuration of an Active Directory server environment utilizing core Windows Server services. The authors simulate the role of system administrators to build a fully functional demo environment leveraging Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), Active Directory Domain Services (AD DS), and Group Policy Objects (GPO).

The entire environment is built and simulated using VMware Workstation as the virtualization platform, which allows the authors to run multiple virtual machines simultaneously on a single physical host. Two virtual machines are deployed within VMware — one acting as the Windows Server 2022 Domain Controller and the other as the Windows Server 2022 client machine. Both virtual machines are configured using Internal Network mode within VMware to ensure isolated and stable communication within the virtual environment, preventing any interference from external networks.

IP address assignment is automated through DHCP, while DNS handles domain name resolution to enable seamless connectivity between server and client. User accounts and domain resources are managed centrally through AD DS, where clients must authenticate using registered domain accounts. Security and access restrictions are enforced through GPO, which limits client users from accessing certain system features and settings.

The overall objective is to construct a fully operational demo environment within VMware Workstation that simulates real-world Active Directory server administration using essential Windows Server services.

Services

1. **Active Directory Domain Services (AD DS)**

A Microsoft service that centralizes user, computer, and network management in a domain environment, allowing administrators to manage authentication, security, and access control efficiently.

2. **Domain Name System (DNS)**

DNS is a service that translates domain names into IP addresses, helping devices communicate easily within a network environment.

3. **Dynamic Host Configuration Protocol (DHCP)**

DHCP automatically assigns IP addresses and network configurations to devices connected to the network, simplifying network management.

4. **Group Policy Object (GPO)**

A set of centralized rules and configurations used to manage user and computer settings within a Windows domain environment.

IP TABLE

Device	IP Address	Subnet Mask	Default Gateway	DNS Server	Role
Domain Controller (Server)	192.168.10.1	255.255.255.0	None (Directly connected)	192.168.10.1	AD DS, DNS, DHCP, GPO
DHCP Clients	192.168.10.10 – 192.168.10.100 (Range)	255.255.255.0	192.168.10.1	192.168.10.1	DHCP Client

Operating system virtual machine(VMWARE) specification:

1. Windows Server 2022
 - 2GB RAM
 - 60GB Storages
 - 1 CPU
 - 1 Internal Network Adapters (VMnet3)
2. Windows 2022 Client
 - 2GB RAM
 - 60GB Storages
 - 1 CPU
 - 1 Internal Network Adapters (VMnet3)

Do All Installations (Services)

On the Server

Open Server Manager

Manage > Add Roles and Features > Next Until Server Roles then Check DHCP DNS and AD DS > Next until Install

The screenshot shows the 'Add Roles and Features Wizard' window. The title bar reads 'Add Roles and Features Wizard'. The main heading is 'Select server roles'. On the right, it says 'DESTINATION SERVER WIN-HFUNTPBHKTS.network1.com'. A left-hand navigation pane lists steps: 'Before You Begin', 'Installation Type', 'Server Selection', 'Server Roles' (highlighted in blue), 'Features', 'Confirmation', and 'Results'. The main area contains the instruction 'Select one or more roles to install on the selected server.' Below this is a list of roles with checkboxes. The roles listed are: Active Directory Certificate Services (unchecked), Active Directory Domain Services (Installed) (checked), Active Directory Federation Services (unchecked), Active Directory Lightweight Directory Services (unchecked), Active Directory Rights Management Services (unchecked), Device Health Attestation (unchecked), DHCP Server (Installed) (checked), DNS Server (Installed) (checked), Fax Server (unchecked), File and Storage Services (2 of 12 installed) (checked), Host Guardian Service (unchecked), Hyper-V (unchecked), Network Controller (unchecked), Network Policy and Access Services (unchecked), Print and Document Services (unchecked), Remote Access (unchecked), Remote Desktop Services (unchecked), Volume Activation Services (unchecked), Web Server (IIS) (unchecked), and Windows Deployment Services (unchecked). To the right of the list is a 'Description' column with text for 'Active Directory Certificate Services (AD CS)'. At the bottom, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

DESTINATION SERVER
WIN-HFUNTPBHKTS.network1.com

Select one or more roles to install on the selected server.

Roles	Description
☐ Active Directory Certificate Services	Active Directory Certificate Services (AD CS) is used to create certification authorities and related role services that allow you to issue and manage certificates used in a variety of applications.
☒ Active Directory Domain Services (Installed)	
☐ Active Directory Federation Services	
☐ Active Directory Lightweight Directory Services	
☐ Active Directory Rights Management Services	
☐ Device Health Attestation	
☒ DHCP Server (Installed)	
☒ DNS Server (Installed)	
☐ Fax Server	
☒ File and Storage Services (2 of 12 installed)	
☐ Host Guardian Service	
☐ Hyper-V	
☐ Network Controller	
☐ Network Policy and Access Services	
☐ Print and Document Services	
☐ Remote Access	
☐ Remote Desktop Services	
☐ Volume Activation Services	
☐ Web Server (IIS)	
☐ Windows Deployment Services	

< Previous Next > Install Cancel

DHCP Configuration

Installation & Authorization:

1. Server Manager > Manage > Add Roles and Features.
2. Check DHCP Server and click Install.
3. After installation, click the Flag icon > Complete DHCP configuration.
4. Click Commit to authorize the server in Active Directory.

Creating a Scope:

1. Open Tools > DHCP.
2. Expand your server > right-click IPv4 > New Scope.
3. Name: LAN_Scope.
4. Start IP: 192.168.10.10, End IP: 192.168.10.100.
5. Add Exclusions: (Optional) Skip if you don't need to reserve specific IPs within that range.
6. Configure DHCP Options: Select "Yes, I want to configure these options now".
 - o 006 DNS Servers: Ensure your server IP (192.168.10.1) is listed.
7. Click Activate Scope and Finish.

DHCP Configuration

DHCP Configuration

DHCP

File Action View Help

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DHCP	Start IP Address	End IP Address	Description
win-hfuntpbhks.network1.com	192.168.10.10	192.168.10.100	Address range for distribution

win-hfuntpbhks.network1.com

- IPv4
 - Server Options
 - Scope [192.168.10.0] Lan Scope
 - Address Pool
 - Address Leases
 - Reservations
 - Scope Options
 - Policies
 - Policies
 - Filters
 - IPv6

DHCP

File Action View Help

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DHCP	Client IP Address	Name	Lease Expiration	Type	U
win-hfuntpbhks.network1.com	192.168.10.10	CLIENT.network1.c...	5/1/2026 1:44:14 AM	DHCP	00

win-hfuntpbhks.network1.com

- IPv4
 - Server Options
 - Scope [192.168.10.0] Lan Scope
 - Address Pool
 - Address Leases
 - Reservations
 - Scope Options
 - Policies
 - Policies
 - Filters
 - IPv6

DNS Configuration

DNS translates names (like *google.com*) into IP addresses. It is automatically installed with AD DS, but needs two final touches.

A. Set Up Forwarders (For Internet Access)

1. Open Tools > DNS.
2. Right-click your Server Name > Properties > Forwarders tab.
3. Click Edit and type 8.8.8.8 (Google's DNS). This allows your local users to browse the internet.

B. Create a Reverse Lookup Zone

1. In DNS Manager, right-click Reverse Lookup Zones > New Zone.
2. Follow the wizard: Primary Zone > To all DNS servers in this domain > IPv4.
3. Network ID: Enter the first three parts of your IP: 192.168.10.
4. Click Finish. This allows the server to identify a computer's name just by its IP address.

DNS Configuration

DNS Configuration

DNS Manager

File Action View Help

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DNS	Name	Type	Data
WIN-HFUNTPBHKTS	(same as parent folder)	Start of Authority (SOA)	[4], win-hfuntpbhks, hos...
Forward Lookup Zones	(same as parent folder)	Name Server (NS)	win-hfuntpbhks.
> _msdcs.network1.com	network1	Host (A)	192.168.10.1
> com			
> network1.com			
> Reverse Lookup Zones			
> Trust Points			
> Conditional Forwarders			

DNS Manager

File Action View Help

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DNS	Name	Type	Data
WIN-HFUNTPBHKTS	(same as parent folder)	Start of Authority (SOA)	[4], win-hfuntpbhks, hos...
Forward Lookup Zones	(same as parent folder)	Name Server (NS)	win-hfuntpbhks.
> _msdcs.network1.com	192.168.10.1	Pointer (PTR)	network.
> com	192.168.10.10	Pointer (PTR)	CLIENT.network1.com.
> network1.com			
Reverse Lookup Zones			
> 10.168.192.in-addr.ar			
> Trust Points			
> Conditional Forwarders			

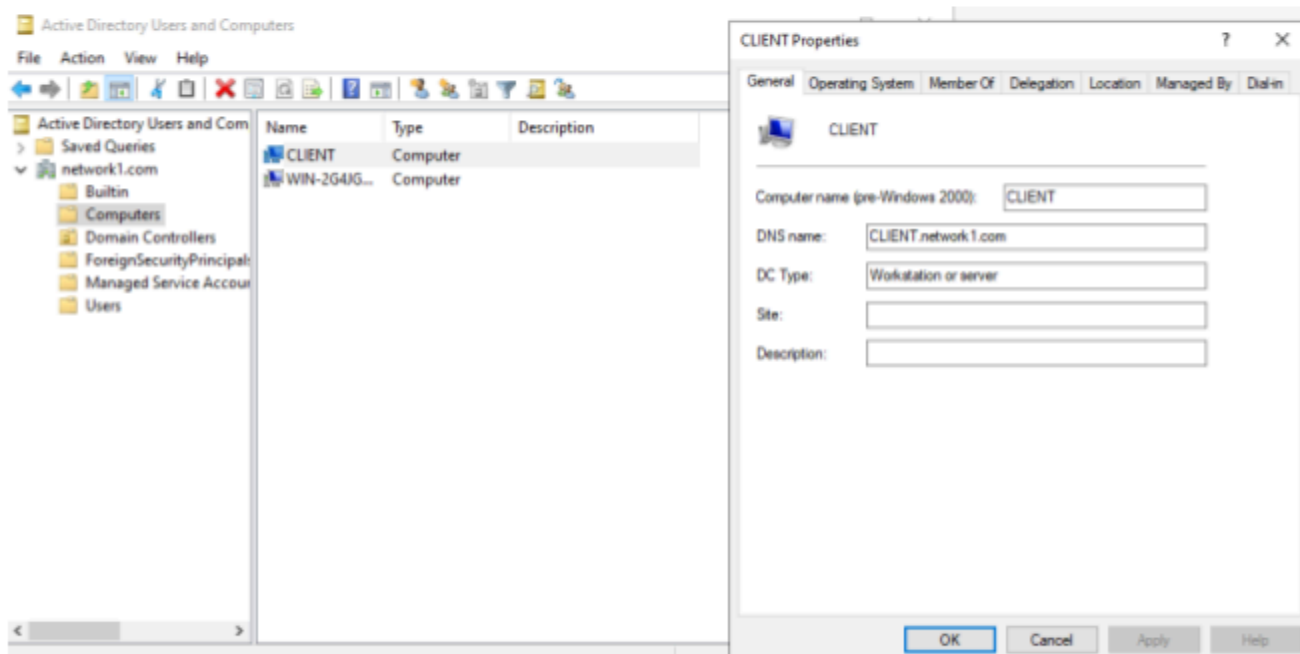
AD DS Configuration Server & Client

A. Installing the Role

1. In Server Manager, click Manage > Add Roles and Features.
2. Click Next until you reach Server Roles.
3. Check Active Directory Domain Services. In the pop-up, click Add Features.
4. Click Next through the rest of the wizard and click Install.

B. Promoting the Server to a Domain Controller

1. Once installation is finished, click the Notifications (Flag icon) in the top bar.
2. Select "Promote this server to a domain controller".
3. Choose "Add a new forest".
4. Root domain name: Type network1.com.
5. DSRM Password: Create a password (you will need this if the database ever crashes).
6. Click Next through the options (the system will handle DNS and NetBIOS).
7. Click Install. The server will automatically restart.



AD DS Configuration Server & Client

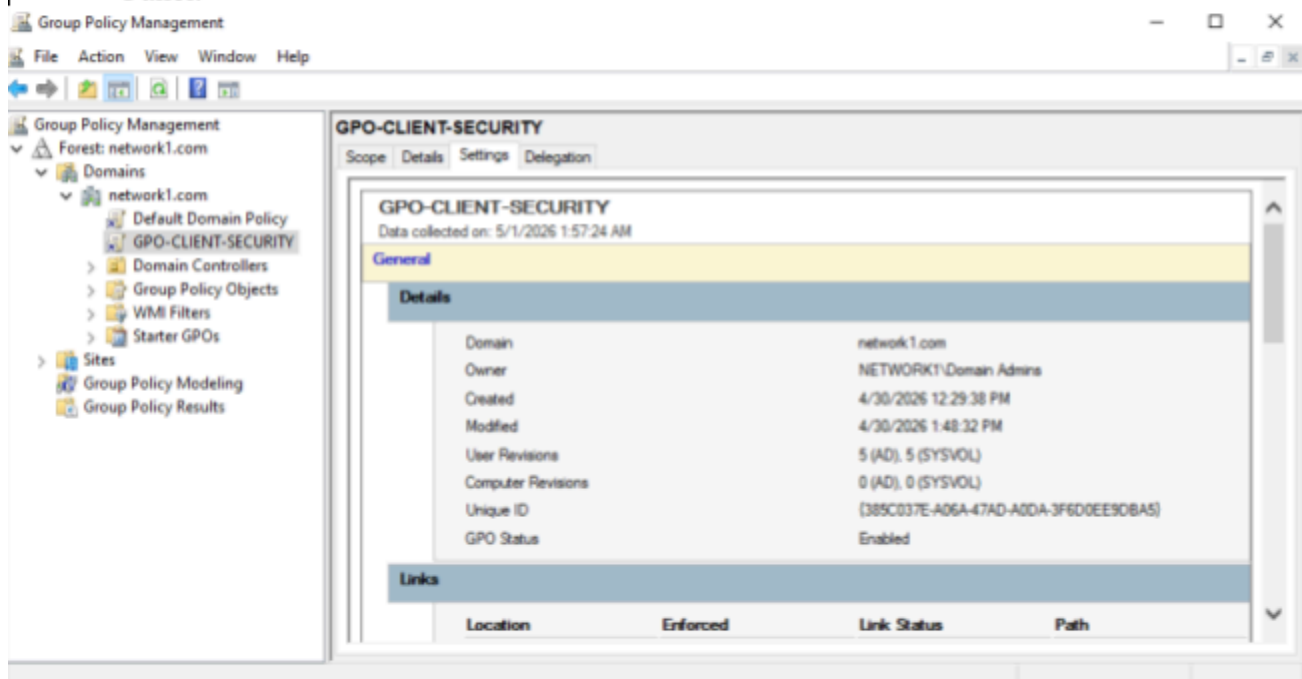
Connect the Client to the Domain

1. On the CLIENT computer, ensure the network settings are set to "Obtain an IP address automatically".
2. Open System Properties > Computer Name tab > Change.
3. Select Domain and type network1.com.
4. Enter the server's Administrator username and password.
5. Restart the client. Success Check:
 - On the Server, open Active Directory Users and Computers. The computer CLIENT will be visible in the Computers folder.
 - In the DHCP console, you will see 192.168.10.10 assigned to the client under Address Leases.

GPO Configuration

If you have installed AD DS, GPO is included in it
GPOs are rules that apply to everyone in the domain.

1. Open Tools > Group Policy Management.
2. Expand your forest > Domains > network1.com.
3. Right-click network1.com > Create a GPO in this domain, and Link it here....
4. Name: GPO-CLIENT-SECURITY.
5. Right-click your new GPO > Edit.
6. Example Policy: Go to *User Configuration > Policies > Administrative Templates > Control Panel*.
 - Enable "Prohibit access to Control Panel".
7. Now, any user who logs in to a computer in this domain will be unable to open the Control Panel.



Testing DHCP on the Client

In terminal run “ping 192.168.10.1” (Ip Server)

```
C:\Users\Administrator.NETWORK1>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=2ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

In terminal run “ping 192.168.10.10” (Ip Client)

```
C:\Users\Administrator.NETWORK1>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Testing DHCP on the Client

In terminal run “ipconfig /all”

```
Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . . . : 
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-CF-D4-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b980:529b:cb5c:477%5(Preferred)
    IPv4 Address. . . . . : 192.168.10.10(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Friday, May 1, 2026 9:45:49 AM
    Lease Expires . . . . . : Wednesday, May 6, 2026 7:53:21 AM
    Default Gateway . . . . . : 
    DHCP Server . . . . . : 192.168.10.1
    DHCPv6 IAID . . . . . : 100666409
    DHCPv6 Client DUID. . . . . : 00-01-00-01-31-86-08-3B-00-0C-29-CF-D4-3A
    DNS Servers . . . . . : 192.168.10.1
                           192.168.23.2
    Primary WINS Server . . . . . : 192.168.23.2
    NetBIOS over Tcpip. . . . . : Enabled
```

Make sure DCHP Enabled : Yes and the IP matches what we set

Testing DNS on the Client

[network1.com](#) = 192.168.10.1

This is according to our DNS server settings

In terminal run “nslookup [network1.com](#)”

```
C:\Users\Administrator.NETWORK1>nslookup network1.com
Server: network
Address: 192.168.10.1

Name: network1.com
Addresses: 192.168.10.1
```

In terminal run “ping [network1.com](#)”

```
C:\Users\Administrator.NETWORK1>ping network1.com

Pinging network1.com [192.168.10.1] with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=4ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=3ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms
```

Testing AD DS on the Client

Test AD DS (Active Directory)

Check the connection to the Domain Controller:

Check if the client can authenticate to AD:

In terminal run “nslookup 192.168.10.10”

```
C:\Users\Administrator.NETWORK1>nslookup 192.168.10.10
Server: network
Address: 192.168.10.1

Name: CLIENT.network1.com
Address: 192.168.10.10
```

Focus on CLIENT.network1.com

Then in terminal run “ping CLIENT network1.com”

```
C:\Users\Administrator.NETWORK1>ping CLIENT.network1.com

Pinging CLIENT.network1.com [fe80::286a:7fcb:32da:7b31%2] with 32 bytes of data:
Reply from fe80::286a:7fcb:32da:7b31%2: time<1ms
Reply from fe80::286a:7fcb:32da:7b31%2: time<1ms
Reply from fe80::286a:7fcb:32da:7b31%2: time<1ms
Reply from fe80::286a:7fcb:32da:7b31%2: time<1ms

Ping statistics for fe80::286a:7fcb:32da:7b31%2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

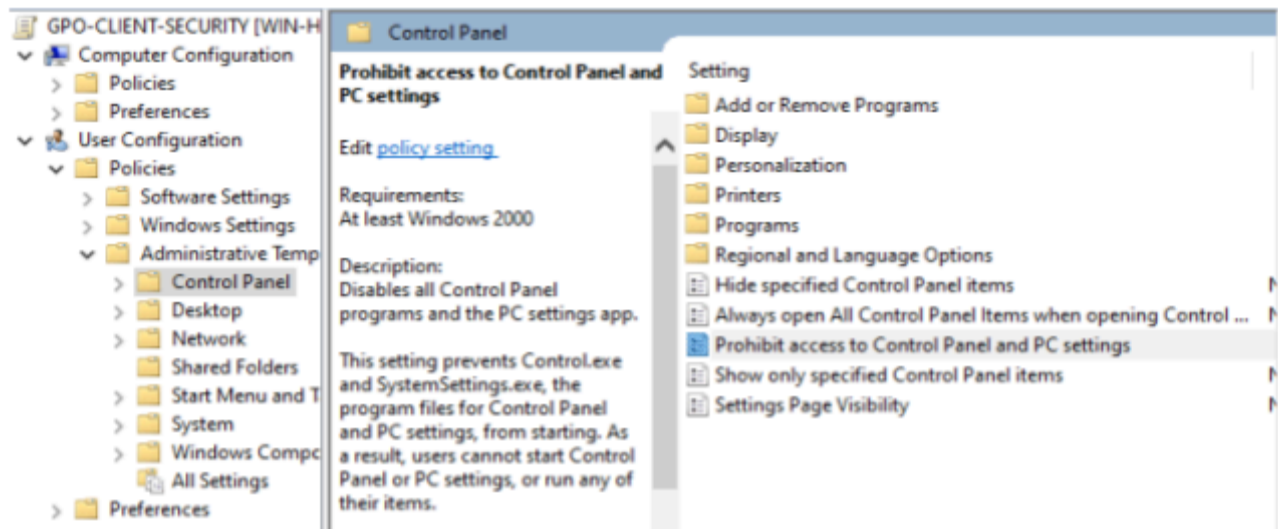
Testing GPO on the Server & Client

In this case, we will use an example, not providing access to the control panel on the client.

On the Server

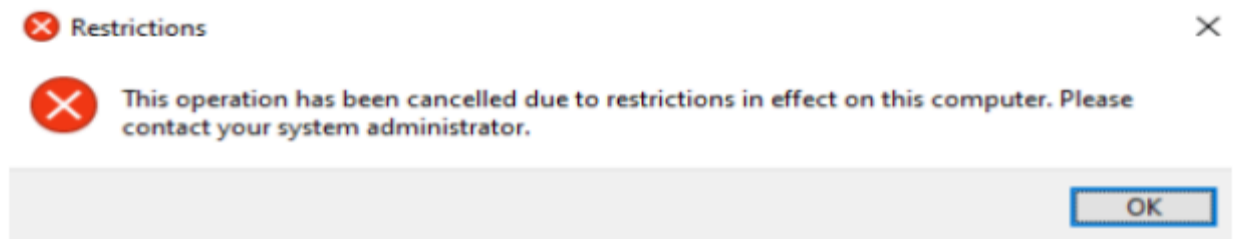
In terminal run “gpmc.msc”

1. User Configuration
2. Policies
3. Administrative Templates
4. Control Panel
5. Prohibit access to Control Panel and PC Settings (Click 2 times)
6. Enable then Click Ok



On the Client

Open Control Panel



If an error occurs, it means the GPO was successful.

This is the display when client tries to open control panel.

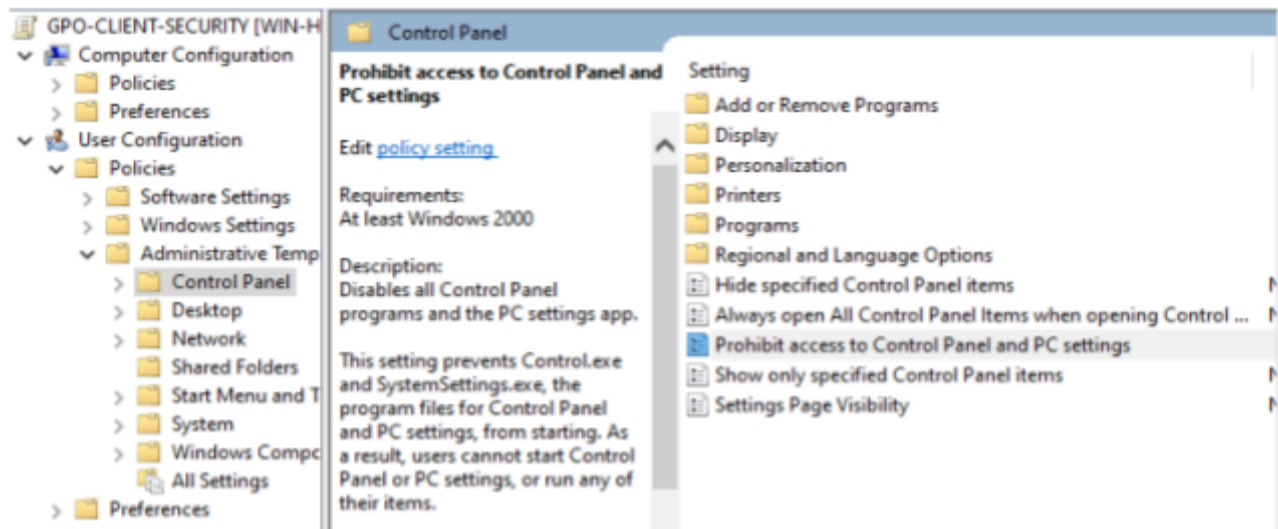
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On the Server

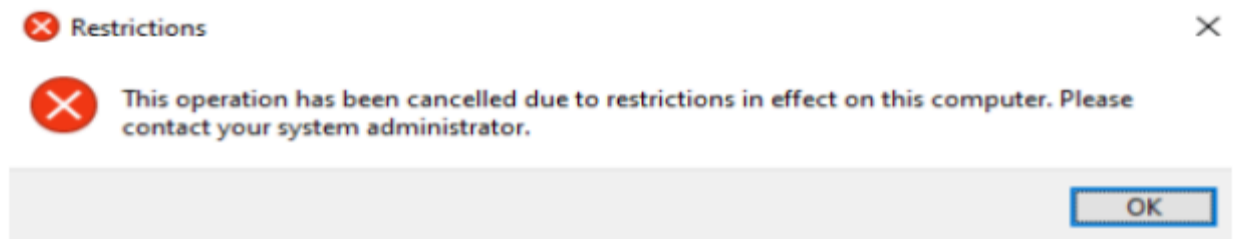
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On the Client

Open Control Panel



If an error occurs, it means the GPO was successful.

This is the display when client tries to open control panel.

Configuring DNSSEC

Open Server Manager > Tools > Win-HFUNTPBHKTS > Forward Lookup Zones > The domain you have set, in here ("network1.com") right click > DNSSEC > Sign the zone > Select use default settings to sign the zone > next until finish.

Then Validation the Sign in Using Powershell
this is the result

```
PS C:\Users\Administrator> Resolve-DnsName network1.com -server 192.168.10.1 -dnssecok
```

Name	Type	TTL	Section	IPAddress
-----	----	---	-----	-----
network1.com	A	600	Answer	192.168.23.149
network1.com	A	600	Answer	192.168.10.1


```
Name       : network1.com
QueryType  : RRSIG
TTL        : 600
Section    : Answer
TypeCovered : A
Algorithm  : 8
LabelCount : 2
OriginalTtl : 600
Expiration : 5/23/2026 9:25:55 PM
Signed     : 5/13/2026 8:25:55 PM
Signer     : network1.com
Signature  : {150, 6, 141, 195...}
```



```
Name       : .
QueryType  : OPT
TTL        : 32768
Section    : Additional
Data       : {}
```

Configuring Socket Pool

Configure the socket pool to change the size to 3000.

a) On the taskbar, select the Windows PowerShell button.

b) In the PowerShell window, enter the following to view the socket pool size:

```
dnscmd /info /socketpoolsize
```

c) In the PowerShell window, enter the following to modify the socket pool size value:

```
dnscmd /config /socketpoolsize 3000
```

Caution: Ensure that you include the value for socketpoolsize along with the command. If you do not enter the value in the above command, the socketpoolsize will be assigned the value 0, which will make the DNS server vulnerable.

d) In the PowerShell window, enter the following to verify the changes:

```
dnscmd /info /socketpoolsize
```

```
PS C:\Users\Administrator> dnscmd /info /socketpoolsize
```

```
Query result:
```

```
Dword: 2500 (000009C4)
```

```
Command completed successfully.
```

```
PS C:\Users\Administrator> dnscmd /config /socketpoolsize 3000
```

```
Registry property socketpoolsize successfully reset.
```

```
Command completed successfully.
```

```
PS C:\Users\Administrator> dnscmd /info /socketpoolsize
```

```
Query result:
```

```
Dword: 3000 (00000BB8)
```

```
Command completed successfully.
```

Configuring Socket Pool

Change the values for the excludedportrange parameter.

a) In the PowerShell window, enter the following to view the current value of the excludedportranges parameter:

```
dnscmd /info /socketpoolexcludedportranges
```

Note: The query result section is blank in the command output because there is no default value.

b) In the PowerShell window, enter the following to change the excluded port ranges:

```
dnscmd /config /socketpoolexcludedportranges 45000-50000
```

c) In the PowerShell window, enter the following to verify the changes:

```
dnscmd /info /socketpoolexcludedportranges
```

```
PS C:\Users\Administrator> dnscmd /info /socketpoolexcludedportranges
Query result:
Command completed successfully.

PS C:\Users\Administrator> dnscmd /config /socketpoolexcludedportranges 45000-50000
Registry property socketpoolexcludedportranges successfully reset.
Command completed successfully.

PS C:\Users\Administrator> dnscmd /info /socketpoolexcludedportranges
Query result:
String: 45000-50000
Command completed successfully.
```

Configuring Cache Locking

Configure the cache locking percentage.

a) Enter the following to view the percentage:

```
dnscmd /info /cachelockingpercent
```

b) Enter the following to modify the value:

```
dnscmd /config /cachelockingpercent 50
```

c) Enter the following to view the value again:

```
dnscmd /info /cachelockingpercent
```

```
PS C:\Users\Administrator> dnscmd /info /cachelockingpercent

Query result:
Dword: 100 (00000064)

Command completed successfully.

PS C:\Users\Administrator> dnscmd /config /cachelockingpercent 50

Registry property cachelockingpercent successfully reset.
Command completed successfully.

PS C:\Users\Administrator> dnscmd /info /cachelockingpercent

Query result:
Dword: 50 (00000032)

Command completed successfully.
```

Configuring IPAM

Join server 2's IPAM domain to server 1

1. Open Server 2 (192.168.10.2)

2. Run Sysprep

Open Run (Win+R) and type:

C:\Windows\System32\Sysprep\sysprep.exe

3. Settings in the Sysprep window

Options

System Cleanup Action | Select Enter System Out-of-Box Experience (OOBE)

Generalize | Checked

Shutdown Options | Reboot

Click OK, and the machine will restart automatically.

4. After rebooting, Server 2 will ask for setup (name, password, etc.). This is normal because a new SID has been generated.

5. Reset Server 2's IP

After setup, set the IP back to 192.168.10.2 because Sysprep can reset the network configuration.

6. Join the domain to Server 1

Go to System Properties > Change > Domain, type your domain name, then enter the admin credentials for Server 1.

System Preparation Tool 3.14



System Preparation Tool (Sysprep) prepares the machine for hardware independence and cleanup.

System Cleanup Action

Enter System Out-of-Box Experience (OOBE) ▼

☐ Generalize

Shutdown Options

Reboot ▼

OK

Cancel

Configuring IPAM

IPAM Setup on Server 2

1. Step 1: Install the IPAM Role
2. On Server 2, open Server Manager:
3. Click Manage > Add Roles and Features
4. Choose Role-based or feature-based installation
5. Select Server 2 (192.168.10.2)
6. In the Features section, check IP Address Management (IPAM) Server
7. Click Install, wait for it to complete, then restart

Next Steps: IPAM Provision

1. Open IPAM in Server Manager
On Server 2, click IPAM in the left pane of Server Manager.

2. Click "Provision the IPAM server."

A wizard will appear. Follow these steps:

Settings

Options

Provisioning method

Group Policy Based

GPO Name Prefix

Type IPAM

Click Apply and wait for the process to complete.

IPAM SERVER TASKS

QUICK START ACTIONS LEARN MORE	1	Connect to IPAM server	4	Start server discovery
	2	Provision the IPAM server	5	Select or add servers to manage
	3	Configure server discovery	6	Retrieve data from managed serv

Configuring IPAM

3. Create GPO on Server 2

Then run

“Invoke-IPAMGpoProvisioning -Domain [network1.com](#)

-GpoPrefixName IPAM -IpamServerFqdn WIN-TKCC2TGJ3G8.[network1.com](#)

-DelegatedGpoUser Administrator”

Yes, type Y for all questions that appear.

Both questions are just confirmations that you agree to the GPO being created and the DNS ACL being modified. Both must indeed be confirmed with Y.

Next Step: Server Discovery

Return to Server 2, open Server Manager > IPAM, then:

Click Configure Server Discovery

Click Add, enter the domain [network1.com](#)

Select the server to manage:

Check Domain Controller

Check DHCP

Check DNS

Click OK

Click Start Server Discovery

Wait for the discovery process to finish

Configuring IPAM

Configure Server Discovery

Select the forest:
network1.com Get forests

Select domains to discover:
Add

Select the server roles to discover:

Domain	Domain controller	DHCP server	DNS server
(root domain) network1.com	☒	☒	☒

Remove

Information For Group Policy based provisioning, create the GPOs for each domain in the list using the Windows PowerShell cmdlet "Invoke-IpamGpoProvisioning" on IPAM server.
[Learn more about Group Policy based provisioning.](#)

Details of server discovery schedule
Next scheduled run time: 5/24/2026 4:49:49 AM
Warning Discovery schedule can be changed by editing \Microsoft\Windows\IPAM\ServerDiscovery located in the Task Scheduler on the IPAM server with administrator privileges.

OK Cancel

Configuring IPAM

IPAM Configuration Steps

Step 1: Connect to IPAM server - Completed

Step 2: Provision the IPAM server - Completed

Step 3: Configure server discovery - Completed (1 domain)

Step 4: Start server discovery

- Click the **"Start server discovery"** text in the upper-right corner of the dashboard.
- This process will automatically search for Server 1 (DC, DHCP, DNS). Wait for it to complete; this may take a few minutes.

Step 5: Select or add servers to manage

- Click **"Select or add servers to manage"**, or navigate via the left panel and click **SERVER INVENTORY**.
- Server 1 will appear there. You need to:
 - Right-click **Server 1**
 - Select **Edit Server**
 - Change the **Manageability Status** from *Unspecified* to *Managed*
 - Click **OK**

Configuring IPAM

If you encounter an error, this may be the cause.

gpupdate has run (discovery timestamp changed), but the status is still Blocked. This is usually because the GPO hasn't been properly applied to Server 1.

Check on Server 1 PowerShell

powershell

```
Get-GPO -Name "IPAM_DHCP"
```

```
Get-GPO -Name "IPAM_DNS"
```

Possible results:

If both are found, it means all GPOs are present. The Blocked issue is with the firewall. Run on the Server 1:

powershell

```
netsh advfirewall firewall set rule group="Remote Event Log Management" new enable=yes
```

```
netsh advfirewall firewall set rule group="Remote Service Management" new enable=yes
```

```
netsh advfirewall firewall set rule group="Windows Management Instrumentation (WMI)" new  
enable=yes
```

If it is not found, it means the GPO has not been created properly, you need to rerun it.

Invoke-IpamGpoProvisioning on Server 2.

Configuring IPAM

All three GPOs already exist:

- IPAM_DHCP: **There is**
- IPAM_DNS: **There is**
- IPAM_DC_NPS: **There is**

This means the Blocked issue is with the firewall. Run the following command on Server 1 PowerShell:

powershell

```
netsh advfirewall firewall set rule group="Remote Event Log Management" new enable=yes  
netsh advfirewall firewall set rule group="Remote Service Management" new enable=yes  
netsh advfirewall firewall set rule group="Windows Management Instrumentation (WMI)" new  
enable=yes  
netsh advfirewall firewall set rule group="Remote Scheduled Tasks Management" new enable=yes  
netsh advfirewall firewall set rule group="Windows Firewall Remote Management" new enable=yes
```

After that run:

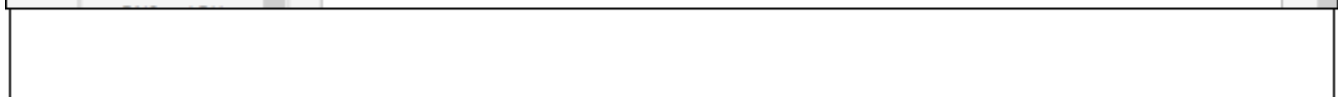
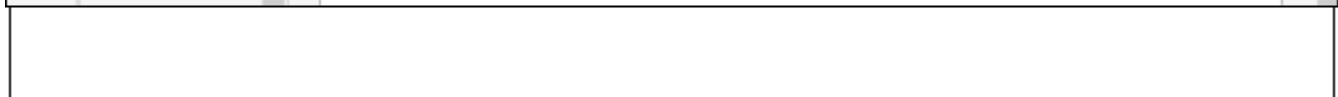
powershell

```
gpupdate /force
```

Then go back to Server 2, right-click the server in SERVER INVENTORY, select Refresh Server Access Status.

If, for example, the three GPOs have been found and the firewall is running, try restarting server 2 (IPAM) directly. Usually, after restarting IPAM will run successfully without errors.

Configuring IPAM



Configuring Recovery

On Server 2, open Server Manager:
Click Manage > Add Roles and Features
Click Next until you reach the Features page
Check Windows Server Backup
Click Install
Wait for it to complete; no restart required

If your RAM can handle it, run this:

The next step:
Create a Backup First
Before recovery, we need to create a backup as a safe point.
Click Local Backup in the left panel
In the right Actions panel, click Backup Once
Select Different options, click Next
Select Custom, click Next
Click Add Items, check Local disk (C:) or a specific folder
Click Next
Select Local drives as the destination
Select an available drive (if there is a D: drive or another volume)
Click Backup

If it can't handle it, try this instead:

Or if you want it simpler, create a dummy file to back up first:

Open PowerShell on Server 2:

powershell

`New-Item -Path "C:\TestBackup" -ItemType Directory`

`New-Item -Path "C:\TestBackup\test.txt" -ItemType File -Value "Ini file test untuk backup"`

Configuring Recovery

On Server 2 PowerShell, create the destination folder:

PowerShell

```
New-Item -Path "C:\BackupDestination" -ItemType Directory
```

Then repeat **Backup Once**. On the **Specify Destination Type** page, select **Remote shared folder**, and enter the path:

`\\WIN-TKCC2TGJ3G8\C$\BackupDestination`

Backup Schedule

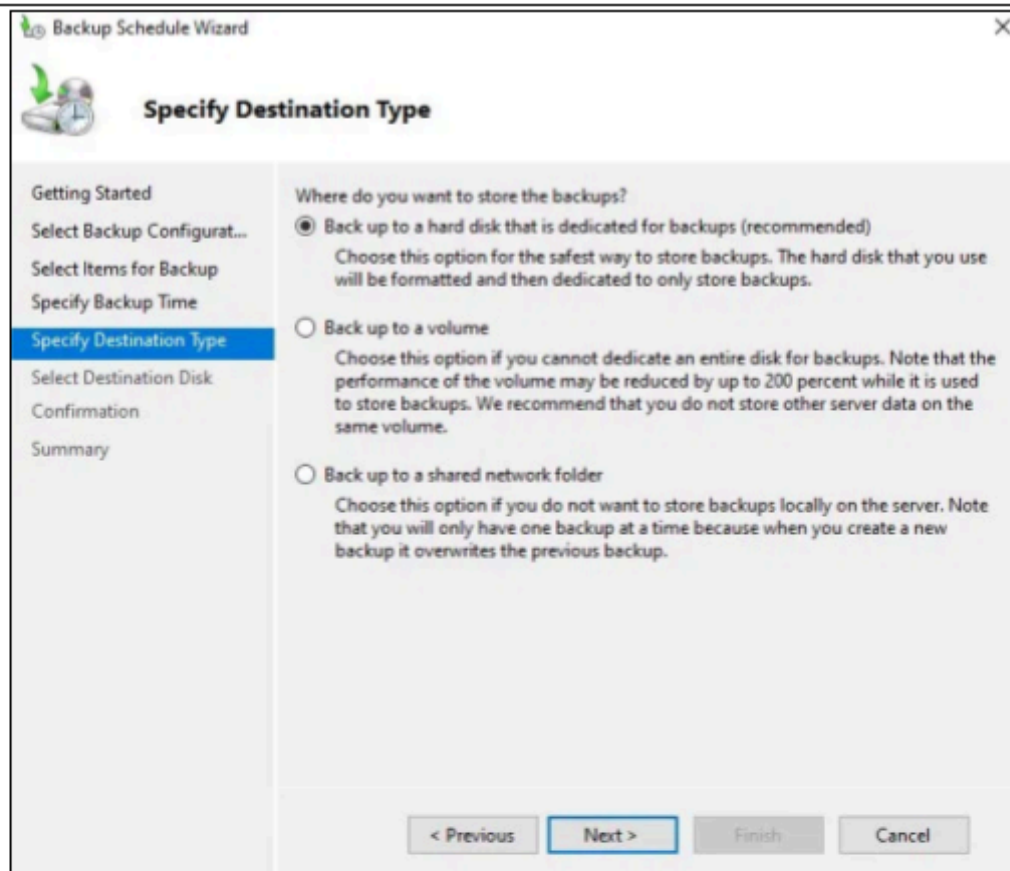
How to schedule a backup:

1. Open **Windows Server Backup**.
2. Click **Backup Schedule**.
3. Configure the daily/weekly schedule.
4. Done, you can take a screenshot immediately for documentation purposes. It's quick, taking only about 5 minutes.

Settings:

- Select **Once a day**.
- Set the time to **9:00 PM** (or any time you prefer).
- Click **Next**.
- Proceed until **Finish**.

Configuring Recovery



This is an example of a schedule backup

Brief explanation:

Option 1 (dedicated disk): requires a separate disk, which your VM doesn't have.

Option 2 (volume): uses an existing partition, which is the most suitable.

Option 3 (network folder): possible, but more complex.

Configuring Recovery

Backup Schedule Wizard

Specify Backup Time

Modify Scheduled Backu...
Select Backup Configurat...
Select Items for Backup
Specify Backup Time
Specify Destination Type
Specify Remote Shared F...
Confirmation
Summary

How often and when do you want to run backups?

☒ Once a day
Select time of day: 12:30 PM

☐ More than once a day
Click an available time and then click Add to add it to the backup schedule.

Available time: 12:00 AM, 12:30 AM, 1:00 AM, 1:30 AM, 2:00 AM, 2:30 AM, 3:00 AM, 3:30 AM, 4:00 AM, 4:30 AM

Scheduled time: 12:30 PM

Add > < Remove

< Previous Next > Finish Cancel

Status

Last Backup	Next Backup	AI
Status: Successful	Status: Scheduled	To
Time: 5/29/2026 12:21 AM	Time: 5/29/2026 12:30 PM	La
View details	View details	OI

Scheduled Backup

A regular scheduled backup is configured for this server

Settings	Destination usage
Backup items: Selected files (Local disk (C:))	Name: \\192.168.1
File excluded: None	Capacity: Details are
Advanced option: VSS Copy Backup	Used space: Details are
Destination: \\192.168.10.1\CS\BackupFromServer2 (Remote shared fol...	Backups available: Details are
Backup time: Every day 12:30 PM	View details
	Refresh information

Configuring Hyper V

If your laptop supports it, run this:

Fix: Enable Nested Virtualization in VMware

1. Shut down Server 2 first (shut down, do not pause).
2. In VMware, right-click the **Server 2 VM**.
3. Select **Settings**.
4. Click the **Processors** tab.
5. Check **Virtualize Intel VT-x/EPT or AMD-V/RVI**.
6. Click **OK**.
7. Turn Server 2 back on.
8. Retry installing Hyper-V.

Open **Server Manager > Add Roles and Features**:

- Click **Next** until you reach the **Server Roles** page.
- Check **Hyper-V**.
- Keep clicking **Next**; skip the **Virtual Switches** page (do not create a new virtual switch so that the network settings remain unchanged).
- Leave the defaults on the **Migration** and **Default Stores** pages.
- Click **Install**.

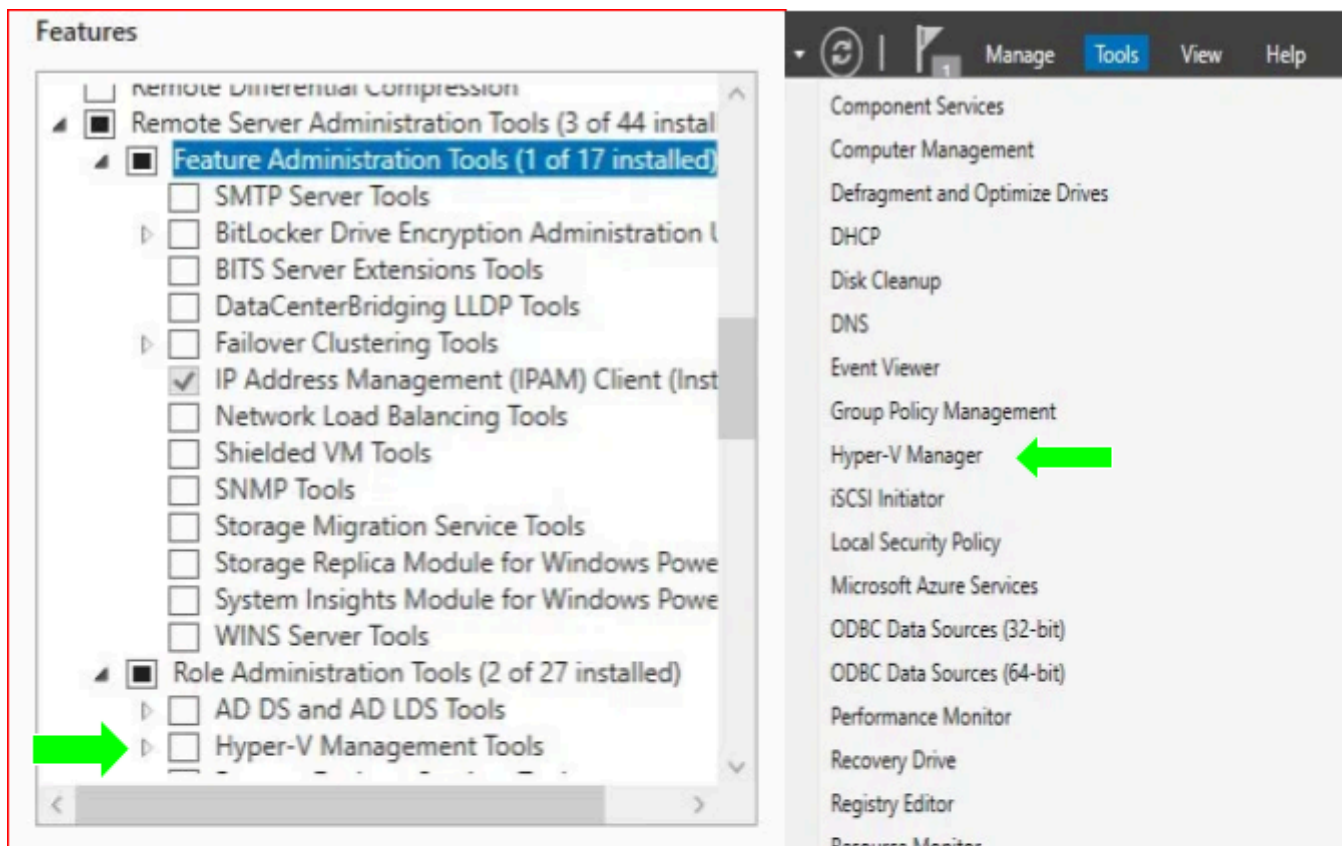
If it is not supported, try installing just the feature instead:

Option 2: Install Hyper-V Management Tools Only

(This is not the full Hyper-V role, but rather tools to manage Hyper-V remotely. This can be installed without nested virtualization)

1. Open **Server Manager > Add Roles and Features**.
2. Go to the **Features** section (not Server Roles).
3. Check **Remote Server Administration Tools > Role Administration Tools > Hyper-V Management Tools**.
4. Click **Install** (no restart required).

Configuring Hyper V



This is the result of a laptop that does not support it, here we only install it without doing any further configuration.

Note :

we just give examples of what we do, otherwise we don't experience it

Requirements

Hardware :

1. Lenovo Thinkpad X1 Carbon

Operating System :

1. Windows Server 2022 64-bit
2. Windows Server 2022 64-bit
3. Windows Client 2022 64-bit

Software :

1. Vmware
2. Google.docs
3. Google Chrome

Project File Details :

<u>No</u>	<u>Files</u>	<u>Remarks</u>
1.	2CS1 Project 3.pdf	Content of the paper
2.	Project 3 Kel 3.pdf	Presentation file